

IoT & Machine Learning course schedule

Jul 13 – Aug 6, 2026 · Lilly Hall of Life Sciences 2102

Class (CA) Pre-lab (PLA) Lab (LA) Project Special

WEEK 1 · JUL 13–17

Day 1 Mon Jul 13

CA

9:00–10:45 AM

Introduction (Prof. Martin Jun) 9:00–9:30

ML1-1: Data Analysis – Handling & Plotting

PLA · PL0

11:15 AM–2:15 PM

PL0 Colabs 1–3 (Setting up Raspberry Pi)

LA · L0

2:15–4:30 PM

L0 Colab 1

Day 2 Tue Jul 14

CA

9:00–10:45 AM

Guest Speaker 9:00–9:50

ML1-2: Data Analysis – Data Indexing

ML1-3: Data Processing

PLA · PL1

11:15 AM–2:15 PM

PL1 Colabs 1–2 (Environmental Sensors)

ISS check-in

2:30–4:40 PM

Purdue ISS · no lab

Day 3 Wed Jul 15

CA

9:00–10:45 AM

Guest Speaker 9:00–9:50

ML2-1: Signal Processing 1 (FFT)

ML2-2: Signal Processing 2 (STFT)

PLA · PL1

11:15 AM–1:15 PM

PL1 Colab 3 (Environmental Sensors)

LA · L1

1:15–3:30 PM

L1 Colabs 1–2

Seminar

4:00 PM

Day 4 Thu Jul 16

CA

9:00–10:45 AM

ML2-3: Signal Processing 3 (WT)

PLA · PL2

11:15 AM–2:15 PM

PL2 Colabs 1–2 (Accelerometer & Signal Processing)

LA · L2

2:15–4:30 PM

L2 Colabs 1–3

Day 5 Fri Jul 17**CA**

9:00–10:45 AM

ML3-1: Feature Extraction

ML3-2: Feature Selection (p-value)

PLA · PL3

11:00 AM–12:00 PM

PL3 Colabs 1–2 (Real-Time Communication)

LA · L3

1:30–3:15 PM

L3 Colabs 1–3

CMSC Tour

3:45 PM–4:30 PM

WEEK 2 · JUL 20–24**Day 6** Mon Jul 20**CA**

9:00–10:45 AM

ML4-1: Dim. Reduction & Visualization 1 (PCA)

ML4-2: Dim. Reduction & Visualization 2 (t-SNE)

PLA · PL4

11:15 AM–2:15 PM

PL4 Colab 1 (Middleware)

LA · L4

2:15–3:30 PM

L4 Colabs 1–3

MMRL Project

3:30–4:30 PM

Machine Tool Monitoring & Data Visualization · Kick-off

MMRL Tour

5:00–5:45 PM

Day 7 Tue Jul 21**CA**

9:00–10:45 AM

ML5-1: Artificial Neural Network (ANN)

PLA · PL5

11:15 AM–2:15 PM

PL5 Colab 1 (MTConnect & HTTP)

LA · L5

2:15–3:30 PM

L5 Colabs 1–3

Industry Tour - Kirby Risk

3:30 - 5:30 PM

Day 8 Wed Jul 22**CA**

9:00–10:45 AM

ML6-1: Convolutional Neural Network (CNN)

ML7-1: Grid Search 1 (ANN)

PLA · PL6

11:00 AM–12:00 PM

PL6 Colab 1 (SQL)

Industry Tour - Evonik

1:30–3:30 PM

Seminar

4:00 PM

Day 9 Thu Jul 23**CA**

9:00–10:45 AM

ML7-2: Grid Search 2 (CNN)

LA · L6

11:00 –12:30 PM

L6 Colabs 1–2

MMRL Project

1:30–4:30 PM

Day 10 Fri Jul 24**MMRL Project**

9:00 AM–12:00 PM

Midterm presentations

1:30 PM–4:30 PM

MMRL Project midterm presentations

WEEK 3 · JUL 27–31**Day 11** Mon Jul 27**CA**

9:00–10:45 AM

ML8: Industrial Data Processing

PLA · PL7

11:00 AM–12:15 PM

PL7 Colab 1 (Grafana)

MSM dataset briefing for Mini Project

LA · L7

1:30–3:30 PM

L7 Colabs 1–2

MMRL Project

3:30–4:30 PM

Day 12 Tue Jul 28**PLA · PL8**

9:00–11:00 AM

PL8 Colabs 1–3 (ML Experiments)

LA · L8

11:00–2:30 PM

L8 Colabs 1–2

PLA · PL9

2:45–4:30 PM

Day 13 Wed Jul 29**LA · L9**

9:00–11:30 AM

MMRL Project

1:00–3:30 PM

PL9 Colab 1 (Implement ML to Raspberry Pi)

then L9 Colabs 1–2

Seminar

4:00 PM

Day 14 Thu Jul 30**PLA · PL10**

9:00 AM - 11: AM

LA · L10

11:15 –4:30 PM

L10 Colabs 1–3

Day 15 Fri Jul 31**Mini Project + MMRL Project**

9:00 AM–4:30 PM

Finish Mini Project MSM · MMRL Project

WEEK 4 · AUG 3–6**Day 16** Mon Aug 3**MMRL Project**

9:00 AM–4:30 PM

Day 17 Tue Aug 4**MMRL Project**

9:00 AM–4:30 PM

Day 18 Wed Aug 5**MMRL Project · final day**

9:00 AM–12:30 PM

Final refinements & presentation
preparation**Seminar**

4:00 PM

Day 19 Thu Aug 6**Final presentations**

9:00 AM–12:00 PM

MMRL Project final presentations ·
course wrap-up